Comparing physician and artificial intelligence chatbot responses to posthysterectomy questions posted to a public social media forum

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BACKGROUND: Within public online forums, patients often seek reassurance and guidance from the community regarding postoperative symptoms and expectations, and when to seek medical assistance. Others are using artificial intelligence in the form of online search engines or chatbots such as ChatGPT or Perplexity. Artificial intelligence chatbot assistants have been growing in popularity; however, clinicians may be hesitant to use them because of concerns about accuracy. The online networking service for medical professionals, Doximity, has expanded its resources to include a Health Insurance Portability and Accountability Act—compliant artificial intelligence writing assistant, Doximity GPT, designed to reduce the administrative burden on clinicians. Health professionals learn using a "medical model," which greatly differs from the "health belief model" that laypeople learn through. This mismatch in learning perspectives likely contributes to a communication mismatch even during digital clinician—patient encounters, especially in patients with limited health literacy during the perioperative period when complications may arise.

OBJECTIVE: This study aimed to evaluate the ability of artificial intelligence chatbot assistants (Doximity GPT, Perplexity, and ChatGPT) to generate quality, accurate, and empathetic responses to postoperative patient queries that are also understandable and actionable.

STUDY DESIGN: Responses to 10 postoperative queries sourced from HysterSisters, a public forum for "woman-to-woman hysterectomy support," were generated using 3 artificial intelligence chatbot assistants (Doximity GPT, Perplexity, and ChatGPT) and a minimally invasive gynecologic surgery fellowship—trained surgeon. Ten physician evaluators compared the blinded responses for quality, accuracy, and empathy. A separate pair of physician evaluators scored the responses for understandability and actionability using the Patient Education Materials Assessment Tool for Printable Materials (PEMAT-P). The final scores were the average of both reviewers' scores. Analysis of variance was used for pairwise comparison of the evaluator scores between sources. Lastly, the Kruskal—Wallis test was used to analyze Flesch—Kincaid scoring for readability. The Pearson chi-square test was used to demonstrate the difference in reading level among the responses for each source.

RESULTS: Compared with a physician, Doximity GPT and ChatGPT were rated as more empathetic than a minimally invasive gynecologic surgeon, but quality and accuracy were similar across these sources. There was a significant difference between Perplexity and the other response sources, favoring the latter, for quality and accuracy ($P<.001$). Perplexity and the minimally invasive gynecologic surgeon ranked similarly for empathy. Reading ease was greater for the minimally invasive gynecologic surgeon responses (60.6 [53.5—68.4]; eighth and ninth grade) than for Perplexity (40.0 [28.6—47.2], college) and ChatGPT (35.5 [28.2—42.0], college) ($P<.01$). There was no significant difference in understandability and actionability, with all sources scored as having good understandability and average actionability.

CONCLUSION: As artificial intelligence chatbot assistants grow in popularity, including integration in the electronic health record, the output's readability must reflect the general population's health literacy to be impactful and effective. This analysis serves as a reminder for physicians to be mindful of this mismatch in readability and general health literacy when considering the integration of artificial intelligence chatbot assistants into patient care. The accuracy and consistency of these chatbots may also impact patient outcomes, making screening of utmost importance in this endeavor.

Key words: artificial intelligence, chatbot assistant, gynecologic surgery, hysterectomy, patient education, postoperative care

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Data availability. The data sets generated and/or analyzed during the current study are available from the corresponding author, S.K.B., upon reasonable request.

Patient consent was not required because no personal information or details were included.

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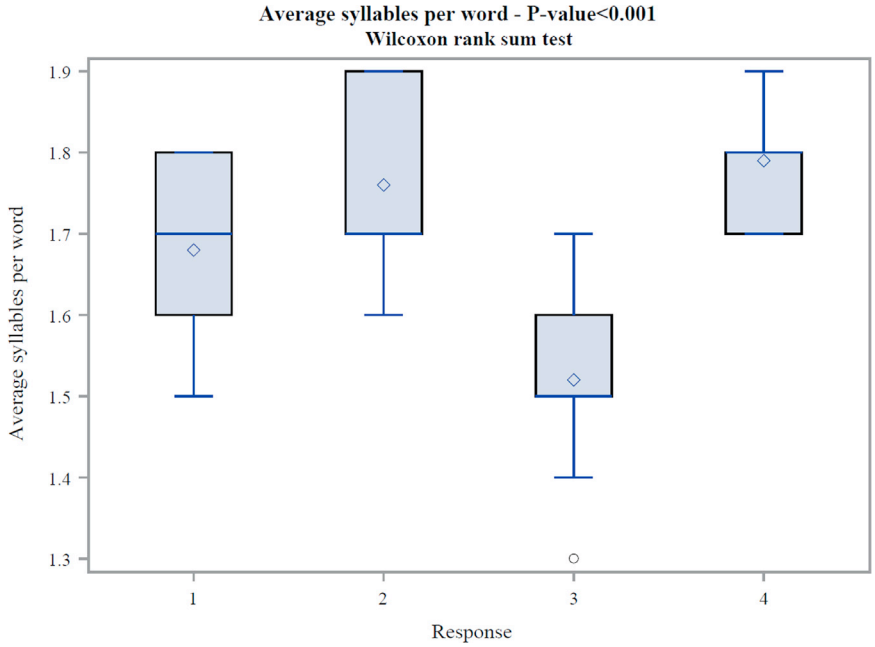
FIGURE 2
Box and whisker plot of average syllables per word



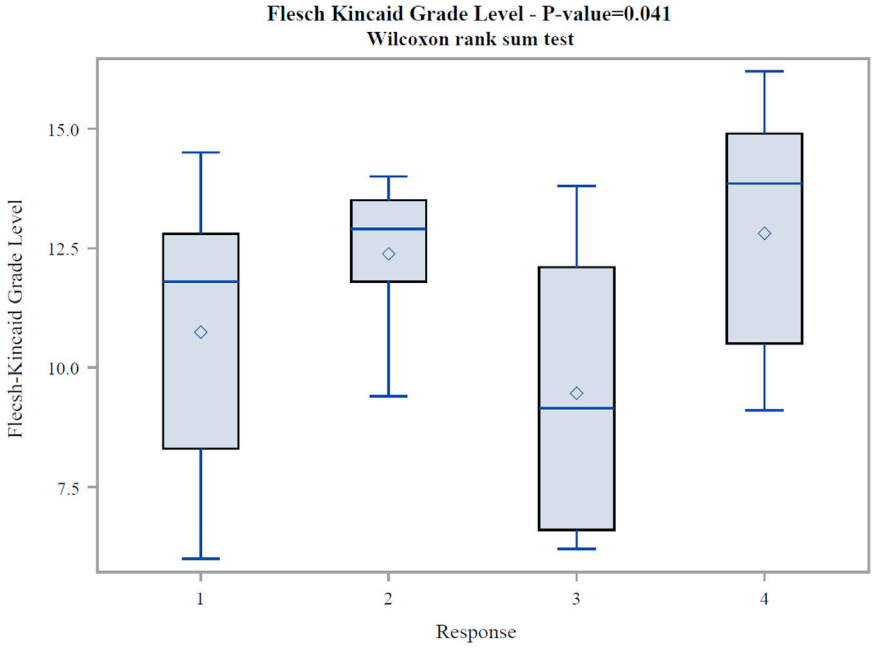
Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

reliability and the differences between the generated responses. Pairwise comparisons were adjusted for multiple testing using the Tukey test.

The Kruskal–Wallis test was used to analyze Flesch–Kincaid scoring for grade level, reading ease, average words per sentence, average syllables per word, and word count.[14] The Pearson chi-square test was used to demonstrate the difference in reading level among the responses for each source.

The significance threshold used was $P<.05$. Statistical analyses were performed in SAS, version 9.4 (SAS Institute Inc, Cary, NC).

FIGURE 3
Box and whisker plot of Flesch–Kincaid grade levels



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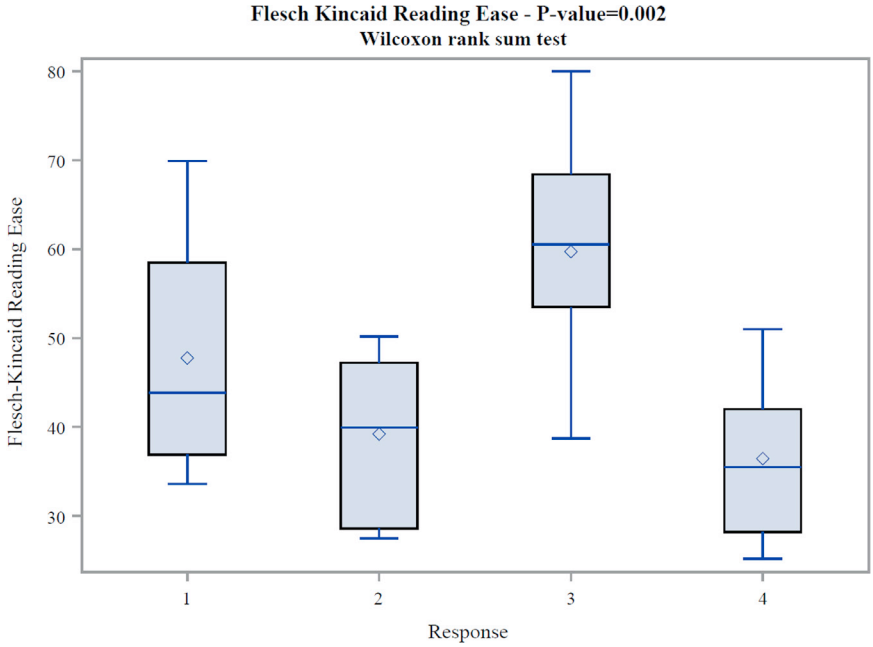
Results

The sample included 10 curated questions intended to reflect common posthysterectomy concerns. Four responses were generated for each question, 1 from each source (Doximity GPT, Perplexity, MIGS surgeon, and ChatGPT), for a total of 40 responses for evaluation. The mean (interquartile range [IQR]) word count for patient questions was 76.9 (58–90), with an average reading ease score of 76.24 (70.88–80.10) corresponding to a seventh-grade reading level. Median (IQR) surgeon responses (90 [72.0–130.0]) were significantly shorter than those generated by Perplexity (236.0 [191.0–255.0]) and ChatGPT (255.5 [192–334]; $P<.001$) (Figure 1). There was no significant difference in the average words per sentence. Surgeon responses

FIGURE 4
Box and whisker plot of Flesch–Kincaid reading ease



Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

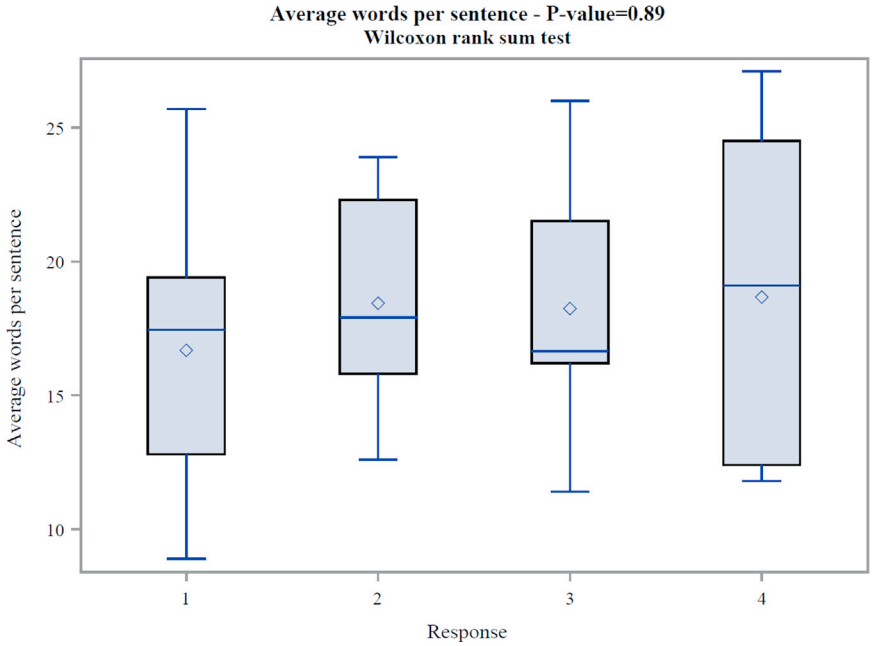
(9.2 [6.6–12.1]) (Figure 2) were at a lower reading grade level than AI chatbot responses, with a significant difference noted for ChatGPT (13.9 [10.5–14.9]; *P*=.04) (Figure 3). Additionally, reading ease was higher for the surgeon responses (60.6 [53.5–68.4], eighth and ninth grade) than for Perplexity (40.0 [28.6–47.2], college) and ChatGPT (35.5 [28.2–42.0], college; *P*<.01) (Figure 4, Table 1).

The evaluators most preferred the Doximity GPT responses, followed by those from ChatGPT, the surgeon, and least of all, Perplexity. No statistically significant difference was noted between the rankings for Doximity GPT and ChatGPT or between the surgeon and ChatGPT responses. There was a significant difference in ranking between Perplexity and all other response sources (*P*<.001) (Tables 2 and 3).

Evaluators rated the quality of responses from Doximity GPT, the surgeon, and ChatGPT highly, with no statistically significant difference noted among these sources. However, a difference was observed between Perplexity and the other sources (*P*<.001), with Perplexity receiving a poorer rating (Tables 4 and 5).

Again, no significant difference was observed in evaluator review for accuracy in the responses from Doximity GPT, the surgeon, and ChatGPT. The average ratings were above *neither disagree nor agree*, leaning more toward *agree*. Although the average accuracy

FIGURE 5
Box and whisker plot of average words per sentence



Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 1
Flesch–Kincaid scoring for physician and artificial intelligence chatbot responses

Question	Response	Median (Q1, Q3)	*P* value	Significance
Flesch–Kincaid grade level	1	11.8 (8.3, 12.8)	.04	3<4
	2	12.9 (11.8, 13.5)		
	3	9.2 (6.6, 12.1)		
	4	13.9 (10.5, 14.9)		
Flesch–Kincaid reading ease	1	43.9 (36.9, 58.5)	<.01	2,4<3
	2	40.0 (28.6, 47.2)		
	3	60.6 (53.5, 68.4)		
	4	35.5 (28.2, 42.0)		
Average words per sentence	1	17.5 (12.8, 19.4)	.89	NS
	2	17.9 (15.8, 22.3)		
	3	16.7 (16.2, 21.5)		
	4	19.1 (12.4, 24.5)		
Average syllables per word	1	1.7 (1.6, 1.8)	<.001	3<1,2<4
	2	1.7 (1.7, 1.9)		
	3	1.5 (1.5, 1.6)		
	4	1.8 (1.7, 1.8)		
Word count	1	159.5 (133.0, 233.0)	<.001	3<2,4
	2	236.0 (191.0, 255.0)		
	3	90.0 (72.0, 130.0)		
	4	255.5 (192.0, 334.0)		

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

assessment for Perplexity was also *neither disagree nor agree*, it leaned more toward *disagree*, and a significant difference was noted between Perplexity and the other sources ($P<.001$) (Tables 6 and 7).

TABLE 2
Average rank of physician and chatbot responses

Response	Estimate (95% CI)
1	2.00 (1.80–2.20)
2	3.21 (3.01–3.41)
3	2.53 (2.33–2.73)
4	2.26 (2.06–2.46)

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

Doximity GPT and ChatGPT responses were similarly rated as more empathetic than Perplexity and surgeon responses. The latter pair was rated as slightly empathetic, whereas the formed was rated as moderately empathetic (Tables 8 and 9).

The evaluators correctly identified that the responses were written by AI 36% of the time for Doximity GPT, 93% of the time for Perplexity, and 84% of the time for ChatGPT. The MIGS surgeon responses were thought to be AI-generated 8% of the time.

The PEMAT-P is a validated tool for assessing the understandability and actionability of printable and audiovisual patient education materials. The printable materials version was used for this study and was scored by 2 evaluators. All responses were considered very short material, defined as a material with ≤2 paragraphs and no more than 1 page in length.[8] Each item was scored as agree (1) or disagree (0), with N/A applying to some items. The final score, expressed as a percentage, was calculated by dividing the sum of the total points by the sum of the total possible points, excluding items scored as N/A, and multiplying by 100. The scores of the 2 evaluators were averaged to obtain the final score. Higher scores indicate greater understandability or actionability of the material.

The responses across sources were of good understandability, with average scores (95% confidence interval) of 76.1% (72.5%–79.6%) for Doximity GPT, 78.9% (75.4%–82.5%) for Perplexity, 75.4% (71.8%–79.0%) for the surgeon, and 77.9% (74.3%–81.5%) for ChatGPT. Although a significant difference in scoring was observed between the 2 evaluators (evaluator 1:81.3% [78.8%–83.9%]; evaluator 2: 72.8% [70.3%–75.3%]; $P<.001$), no significant difference was found among the final scores for the sources (Table 10).

A similar pattern was noted for actionability. There was a difference in evaluator scoring (evaluator 1: 68.2% [62.7%–73.6%]; evaluator 2: 38% [32.5%–43.5%]; $P<.001$), but no significant difference in the final actionability

TABLE 3
Pairwise comparison of physician and chatbot ranks

Compare	Estimate (95% CI)	*P* value
Resp 1 - Resp 2	−1.21 (−1.59 to −0.83)	<.001
Resp 1 - Resp 3	−0.53 (−0.91 to −0.15)	<.01
Resp 1 - Resp 4	−0.26 (−0.64 to 0.12)	.29
Resp 2 - Resp 3	0.68 (0.30−1.06)	<.001
Resp 2 - Resp 4	0.95 (0.57−1.33)	<.001
Resp 3 - Resp 4	0.27 (−0.11 to 0.65)	.26

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval; *Resp*, response.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 4
Average quality rating of physician and chatbot responses

Response	Estimate (95% CI)
1	3.89 (3.71−4.07)
2	2.80 (2.62−2.98)
3	3.60 (3.42−3.78)
4	3.77 (3.59−3.95)

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 5
Pairwise comparison of physician and chatbot quality

Compare	Estimate (95% CI)	*P* value
Resp 1 - Resp 2	1.09 (0.75−1.43)	<.001
Resp 1 - Resp 3	0.29 (−0.05 to 0.63)	.12
Resp 1 - Resp 4	0.12 (−0.22 to 0.46)	.79
Resp 2 - Resp 3	−0.80 (−1.14 to −0.46)	<.001
Resp 2 - Resp 4	−0.97 (−1.31 to −0.63)	<.001
Resp 3 - Resp 4	−0.17 (−0.51 to 0.17)	.56

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval; *Resp*, response.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 6
Average accuracy rating of physician and chatbot responses

Response	estimate (95% CI)
1	3.98 (3.82−4.14)
2	3.06 (2.90−3.22)
3	3.96 (3.80−4.12)
4	3.96 (3.80−4.12)

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 7
Pairwise comparison of physician and chatbot accuracy

Compare	Estimate (95% CI)	*P* value
Resp 1 - Resp 2	0.92 (0.62−1.22)	<.001
Resp 1 - Resp 3	0.02 (−0.28 to 0.32)	1.00
Resp 1 - Resp 4	0.02 (−0.28 to 0.32)	1.00
Resp 2 - Resp 3	−0.90 (−1.20 to −0.60)	<.001
Resp 2 - Resp 4	−0.90 (−1.20 to −0.60)	<.001
Resp 3 - Resp 4	0.00 (−0.30 to 0.30)	1.00

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval; *Resp*, response.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 8
Average empathy rating of physician and chatbot responses

Response	Estimate (95% CI)
1	3.79 (3.59−3.99)
2	2.47 (2.27−2.67)
3	2.58 (2.38−2.78)
4	3.59 (3.39−3.79)

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

scores among the sources (Table 11). The responses were notably average for actionability, with average scores of 55.0% (47.3%−62.7%), 48.3% (40.6%−56.1%), 51.0% (43.3%−58.7%), and 58.0% (50.3%−65.7%) for Doximity GPT, Perplexity, the surgeon, and ChatGPT, respectively.

For both understandability and actionability, there was no difference among the responses, with evaluators rating all sources similarly (Tables 12 and 13).

Comment

Principal findings

In this cross-sectional study examining patient questions about hysterectomy recovery in a public forum, the evaluators preferred the Doximity GPT and ChatGPT responses over those from the surgeon and Perplexity, most notably for empathy. However, there was no significant difference in the ratings for response quality or accuracy among Doximity GPT, the surgeon, and ChatGPT. Perplexity scored lowest in all areas.

For the secondary aim, the surgeon responses had better readability than those of the AI chatbots, with shorter text that more closely matched the demonstrated reading level of the patient responses. The average American adult reads at an eighth-grade level, which is reflected in the patient questions. It is recommended by many organizations, including the American Medical Association and the National Institutes of Health, that patient education materials not exceed a sixth-grade reading level.[15] This study, consistent with many others, demonstrates that AI chatbot output typically corresponds to a college reading level. Despite this discrepancy in readability, all response sources were found to have good understandability with average actionability. Significant disagreement between the evaluators was noted, especially for actionability. Namely, evaluator 1's actionability scores were in the good range, whereas evaluator 2's were poor to average.

Results in the context of what is known

Ayers et al[6] compared physician responses within a public forum with ChatGPT responses, and similarly found that the chatbot was preferred to physicians for quality and empathy. However, our study highlights that this is not universally true across AI chatbot assistants. We demonstrate that the difference between the most studied AI

TABLE 9
Pairwise comparison of physician and chatbot empathy

Compare	Estimate (95% CI)	P value
Resp 1 - Resp 2	1.32 (0.95–1.69)	<.001
Resp 1 - Resp 3	1.21 (0.84–1.58)	<.001
Resp 1 - Resp 4	0.20 (−0.17 to 0.57)	.50
Resp 2 - Resp 3	−0.11 (−0.48 to 0.26)	.87
Resp 2 - Resp 4	−1.12 (−1.49 to −0.75)	<.001
Resp 3 - Resp 4	−1.01 (−1.38 to −0.64)	<.001

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CI, confidence interval.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 10
Average PEMAT-P understandability scores by evaluator

Evaluator	Response	Estimate % (95% CL)
1	1	81.0 (75.9–86.1)
	2	80.7 (75.6–85.7)
	3	79.6 (74.5–84.6)
	4	84.1 (79.0–89.1)
2	1	71.1 (66.1–76.2)
	2	77.2 (72.1–82.3)
	3	71.2 (66.2–76.3)
	4	71.7 (66.6–76.7)

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CL, confidence level; *PEMAT-P*, Patient Education Materials Assessment Tool for Printable Materials.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 11
Average PEMAT-P actionability scores by evaluator

Evaluator	Response	Estimate % (95% CL)
1	1	66.0 (55.1–76.9)
	2	68.7 (57.8–79.6)
	3	64.0 (53.1–74.9)
	4	74.0 (63.1–84.9)
2	1	44.0 (33.1–54.9)
	2	28.0 (17.1–38.9)
	3	38.0 (27.1–48.9)
	4	42.0 (31.1–52.9)

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CL, confidence level; *PEMAT-P*, Patient Education Materials Assessment Tool for Printable Materials.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

chatbot, ChatGPT, and physician responses may not be as vast as previously suggested. Although the length of responses likely factored into the ratings, we demonstrate that physician responses are superior to AI chatbot responses in terms of readability, despite the shorter length.

Prior studies compared AI chatbots with search engines that are increasingly integrating large language models, such as Google and Bing. Accuracy may increase over time, as demonstrated by Pavlik et al[16] in their comparison of virtual assistants, but there remains significant room for improvement. This study provides a different perspective on the understandability and actionability of patient education material, using less formal prompts reflecting the natural language that patients would use.

Clinical implications

AI continues to rise in popularity with the public as a source of information and advice. As the healthcare industry seeks ways to alleviate burnout and moral injury among healthcare providers, AI is becoming an increasingly appealing resource in the management of patient messages in the electronic medical record. In addition to HIPAA-compliance, it is also important to develop a patient-facing AI application that takes readability into account. In the meantime, the knowledge gained from this study can be used to inform more effective use of AI as a tool and better guidance of patients in its application. As misinformation spreads across platforms, this information also becomes a potential resource for AI chatbots to sample. It is important to assess the accuracy of the output generated by common AI chatbots.

Additionally, the results of the PEMAT-P demonstrate that there is room for improvement in both understandability and actionability in responses to patient questions. As AI becomes integrated into almost every aspect of daily life, it seems inevitable that it will be integrated into the electronic medical record. Natural language processing AI assistants learn from their users, which may include

TABLE 12
Pairwise comparison of physician and chatbot understandability scores

Comparison	Estimate % (95% CL)	P value
Eval 1 - Eval 2	8.5 (4.9–12.1)	<.001
Resp 1 - Resp 2	−2.9 (−9.6 to 3.8)	.67
Resp 1 - Resp 3	0.7 (−6.0 to 7.4)	.99
Resp 1 - Resp 4	−1.8 (−8.5 to 4.9)	.89
Resp 2 - Resp 3	3.5 (−3.1 to 10.2)	.51
Resp 2 - Resp 4	1.1 (−5.6 to 7.8)	.97
Resp 3 - Resp 4	−2.5 (−9.2 to 4.2)	.77

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CL, confidence level; *Eval*, Evaluator; *Resp*, response.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

TABLE 13
Pairwise comparison of physician and chatbot actionability scores

Comparison	Estimate % (95% CL)	P value
Eval 1 - Eval 2	30.2 (22.5–37.9)	<.0001
Resp 1 - Resp 2	6.7 (−7.7 to 21.1)	.62
Resp 1 - Resp 3	4.0 (−10.4 to 18.4)	.88
Resp 1 - Resp 4	−3.0 (−17.4 to 11.4)	.95
Resp 2 - Resp 3	−2.7 (−17.1 to 11.7)	.96
Resp 2 - Resp 4	−9.7 (−24.1 to 4.7)	.30
Resp 3 - Resp 4	−7.0 (−21.4 to 7.4)	.58

Response 1, response 2, response 3, and response 4 correspond to Doximity GPT, Perplexity, minimally invasive gynecologic surgeon, and ChatGPT, respectively.

CL, confidence level; *Eval*, evaluation; *Resp*, response.

Beale. Artificial intelligence vs physician: a comparison of posthysterectomy advice. Am J Obstet Gynecol Glob Rep 2025.

incorporating canned responses that many healthcare providers currently use to increase efficiency in responding to common questions. Perhaps this warrants review and updating of existing materials to account for these considerations.

Research implications

This study is among a growing number of studies that include an evaluation of readability when assessing AI assistants. Regardless of the source, the response is not as effective if the patient is unable to comprehend the message. Natural language processing AI assistants, such as those explored in this study, can be prompted to respond at a specific reading level. However, multiple studies, including ours, have demonstrated that this is not an inherent feature of popular AI assistants currently, and responses exceed the recommended reading level of patient education materials.[17,18] We chose to input the patient prompts to reflect how the average user engages with AI chatbots, which does not include requesting a specific reading level. The next step in research needs to include whether chatbots can be trained to generate responses at a particular reading level. Alternatively, as AI is rapidly evolving, a chatbot may be developed that inherently has output at a specific reading level.

Strengths and limitations

The use of online forum questions is both a strength and a limitation of this study. The questions were directed at an audience of nonphysician peers based on shared experience rather than expertise. The phrasing of these questions is similar to messages in the EHR that may be addressed by healthcare providers other than the patient's primary surgeon. Additional context is initially provided that would otherwise be extracted through a series of correspondences or from review of the patient medical record. The responses reflect the impersonal relationship that may exist between the patient and potential respondent. Alternatively, we could not assess the ability of an AI assistant to navigate HIPAA-protected patient queries within an EHR and gather additional context to provide a more personalized response.

Another strength of this study is that there was consistency in the physician response. A prior study evaluated random physician responses in a public forum, with variability in specialty, credentials, and years of experience.[6] Our study used a board-certified, high-volume, MIGS fellowship–trained surgeon with over 15 years of experience and service as a board examiner as an expert in the field. Given that all gynecologic surgeons are expected to share common knowledge on performing hysterectomies, the associated complications, and how to manage common postoperative outcomes, we consider this single surgeon as a suitable and consistent representative of gynecologic surgeons. Additionally, we expanded the evaluation of AI chatbots to include one that is primarily physician-facing with potential to be patient-facing in the future. Although patients are increasingly using AI chatbots to self-triage, physicians may also be using them to ease their mental burden. It is also important to assess AI chatbots for their

potential HIPAA compliance and suitability for use within the EHR. Finally, using blind evaluators, we compared responses from 3 different AI chatbots and a surgeon.

A limitation of this study is the disagreement between the evaluators in scoring the PEMAT-P. Given the objectivity of the assessment tool, it is surprising that there was such a large discrepancy, particularly for actionability. The most likely explanation is a difference in the interpretation of how to assess each item. Although the assessment tool seems straightforward, it is important to carefully read the user's guide, which includes helpful examples of actionable language.

Conclusions

As AI chatbot assistants grow in popularity, including integration in the EHR, the output's readability must reflect the general population's health literacy to be impactful and effective. This analysis serves as a reminder for physicians to be mindful of this mismatch in readability and general health literacy when considering the integration of AI chatbot assistants into patient care. The accuracy and consistency of these chatbots may also impact patient outcomes, making screening of utmost importance in this endeavor. ■

CRediT authorship contribution statement

Shadae K. Beale: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Natalie Cohen:** Conceptualization, Investigation. **Beatrice Secheli:** Investigation. **Donald McIntire:** Formal analysis, Validation, Writing – original draft. **Kimberly A. Kho:** Conceptualization, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.xagr.2025.100553.

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